# Saiteja Venkateshwa Rao Dasari

Robotics Engineer | Controls Engineer | Autonomy Engineer
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## SUMMARY

Robotics engineer with an M.S. in Robotics and Autonomous Systems (ASU, **3.93 GPA**) working across perception, planning, and control. Delivered a **5-robot** ROS2/Gazebo autonomy stack at **99% mission success**, vision-guided manipulation on a **6-DOF** arm, and closed-loop controllers validated from simulation to hardware; strong in Python, C/C++, and MATLAB/Simulink for real-time robotic systems.

## TECHNICAL SKILLS

**Programming & Tools:** Python, C/C++, MATLAB/Simulink, Bash, Linux, Git, SolidWorks.
**Robotics & Simulation:** ROS 1/ROS 2, Gazebo, MuJoCo, CrazySim SITL, RViz, tf2, URDF/xacro, ArduPilot, Mission Planner, cflib.
**Controls & Estimation:** Control systems, PID, closed-loop control, state estimation, sensor fusion, control allocation, fault detection & isolation (FDI), controller reconfiguration, INDI, guidance, navigation & control (GNC).
**Autonomy & Perception:** Motion planning, A*, Hungarian algorithm, costmaps, occupancy grids, collision avoidance, computer vision, OpenCV, YOLOv8 object detection, ArUco pose estimation.
**Test & Hardware:** Sim-to-hardware validation, hardware bring-up and integration, system testing, root cause analysis, flight testing, flight-log analysis.

## EDUCATION

**Arizona State University** Aug 2024 – May 2026
M.S. Robotics and Autonomous Systems, Systems Engineering | GPA: 3.93/4.0 Tempe, AZ
Relevant Coursework: Robotic Systems I and II, Aerial Robotics, Multi-Robot Systems, Mechatronic Systems, Controls and Systems, Linear Algebra.

**Chaitanya Bharathi Institute of Technology** Aug 2020 – May 2024
B.E. Electrical and Electronics Engineering Hyderabad, India
Relevant Coursework: Control Systems, Signals and Systems, Analog Electronics, Digital Electronics, Power Electronics.

## PROJECTS

**Multi-Robot Task Allocation & Navigation** [GitHub] Dec 2025
- Developed a centralized ROS2/Gazebo autonomy stack for **5 TurtleBots** with from-scratch Hungarian assignment, a custom A* path planner (ROS2 service), and an inflation costmap; attained **99% mission success**.
- Built a path cache keyed on start pose and task ID to align allocated and executed routes, cutting redundant A* re-planning.

**Fault-Tolerant Quadrotor Control** [GitHub] Jan 2026 – May 2026
- Derived and validated closed-form 3-motor control allocation for **all 4 single-motor failures**, reaching **0° attitude error across 24 test conditions**; root-caused attitude loss to motor saturation and added a thrust clamp that cut max roll from **145° to 7.2°**.
- Built a MuJoCo rigid-body simulation modeling gyroscopic coupling and validated sim-to-hardware transfer across PID, INDI, and gyroscopic-compensation controllers, testing **9 allocation methods** on hardware.

**Maze Solving & Path Planning** [GitHub] Dec 2024
- Built an OpenCV vision pipeline into a **34×34 occupancy grid**, then computed A* paths and inverse kinematics (MATLAB RigidBodyTree) to execute waypoint trajectories on a physical **6-DOF myCobot Pro 600** arm.

**Autonomous Drone Landing on Moving Platform** [GitHub] May 2025
- Built a Parrot Mambo landing system with HSV segmentation, centroid tracking, velocity-sync logic, and staged PID descent to align with a moving 11×11 in platform; achieved **<5 cm lateral error, ~4 s landing, and >90% success** across varied platform speeds.

## EXPERIENCE

**Honeywell Aerospace** Aug 2025 – Apr 2026
Technical Lead, SkySpeak AI (ASU Collaborative Research) Tempe, AZ
- Directed a **3-engineer team** with Honeywell/ASU stakeholders to deliver an ATC/pilot simulator with LLM agents, guardrails, and a **300 ms** LiveKit/Deepgram/ElevenLabs voice pipeline.
- Engineered Python voice analytics with YIN pitch detection, VAD, and z-score calibration with abstention scoring (<0.60); launched a CBTA dashboard for **20+ stakeholder reviews** and seed-grant support.

**Arizona State University** Jan 2025 – May 2026
Graduate Teaching Assistant, MATLAB Programming, Instrumentation & Controls Lab Tempe, AZ
- Mentored **90+ students** across MATLAB, instrumentation, and controls labs; troubleshot code and hardware setups, led reviews, and reinforced PID and closed-loop control concepts.

**Marut Drones (IIIT-Hyderabad)** Sept 2023 – Mar 2024
Production Engineering Intern, UAV Systems Integration Hyderabad, India
- Assembled, integrated, and verified UAV subsystems (ESCs, flight controllers, GPS, avionics, propulsion), raising production efficiency **15%** across multirotor platforms.
- Root-caused flight-stability issues from ArduPilot/Mission Planner flight logs and tuned PID parameters across **3 UAV variants**, improving flight stability by **~10%**.

## CERTIFICATIONS & PUBLICATION

Universal Robots e-Series Core/Pro | MathWorks AI | Udemy ROS/ROS2 Gazebo | SCADA-Based Substation Control Panel (Publication)